Questions by topic: Ozone depletion - Set 1 (50 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

0 1 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 1 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 1 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 1 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 1 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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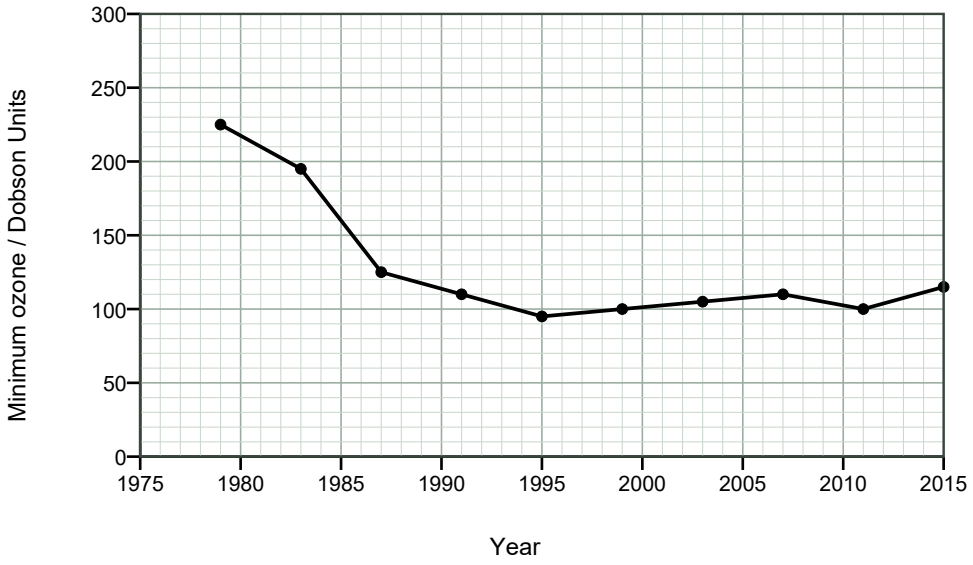
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0 2 Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1



0 2 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 2 . 2 Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]

Answer ..... %

0 2 . 3 Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

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0 2 . 4 Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

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0 3 . 1 Describe the main features of the Montreal Protocol (1987).

[3 marks]

0 3 . 2 For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant .....

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Refrigerant .....

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0 3 . 3 Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

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0 3 . 4 State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

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0 4 . 1 Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

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0 4 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

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0 5 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 5 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 5 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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END OF QUESTIONS

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